

Enhancing Social Connectivity: Tangible Peer-Based Check-in Systems for Isolated Older Adults

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ABSTRACT

Population aging is a global phenomenon. It is expected that by 2050, the older adult population will be 21.4% of the total population in US alone, according to Pew Research. WHO recognizes social engagement as an important factor for reducing isolation. Researchers have historically used digital technologies to build connections, but they offer disadvantages like accessibility issues and have not been a good match for the aged population. My doctoral research aims to (1) understand older adults' needs, preferences, and challenges in maintaining social connections (2) investigate the benefits of tangible technologies to help older adults stay connected (3) create a design framework for tangible technologies to improve social connection for older adults. This doctoral consortium paper discusses my research on exploring the potential of tangible technologies, as opposed to digital, to reduce social isolation among older adults.

CCS CONCEPTS

• Human-centered computing → User interface toolkits.

KEYWORDS

older adults, social isolation, tangible technologies

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1 PROBLEM STATEMENT

1.1 Social Isolation Among Older Adults

Older adults are now the fastest-growing population in the United States [8], they have far-reaching social impacts, and this has led to a growing attention towards aging-related concerns and providing support to the older adults from the HCI research community [21, 30, 37, 46]. About one in four older adults in USA are

considered to be socially isolated [1]. Consequences of social isolation include poorer mental and physical health, and can even lead to premature mortality [28, 35, 36, 60]. While biological processes play an important role, it is also an individual's social and environmental conditions that contribute to aging [18]. Several works have focused on the physical health of older adults, but improving social connection plays an important role in reducing isolation [39]. The Harvard Longitudinal Study of Adult Development [43], which is almost 80 years old and one of the world's longest studies, has showed that embracing community helps us live longer, and happier. The key to healthy aging is close, meaningful relationships [61].

1.2 Independence and Agency in Old Age

Priorities change with age, and more than active engagement, it is important to focus on adapting to life changes while maintaining agency [6]. Independence, in case of older adults, is often guided by agency, which is defined as the ability of an individual to make their own choices and act on them accordingly [57]. However, as people age, they tend to lose agency and others (families, professional caregivers) start making decisions for them. This becomes especially prevalent when older adults are no longer able to perform daily tasks independently. According to the latest AARP survey, older adults express aging in place as a primary goal [64]. HCI researchers have explored various technologies to help older adults age-in-place because older adults want to avoid institutionalization [22]. These technologies include sensor-equipped smart homes [45] and physical in-home systems that share well-being information [55]. However, with aging in place technologies, there is a growing concern that these technologies will actually replace human contact, in-person interactions, and give rise to isolation among older adults [40, 41]. Other technologies include wearables that monitor various health parameters 24x7 [49] and digital location tracking apps in smartphones [56] to track movements. Such passive monitoring takes away agency and control from older adults, which is dehumanizing and undesirable [13, 40].

1.3 Ageism in Technology Design

With the advent of digital technologies, communication has become easy, but screen-based interfaces do not offer an intuitive, accessible medium for older adults resulting in a steeper learning curve [5, 32] and technology adoption difficult. Furthermore, the biometrics or tactile feedback do not work as people age [24], because the firmness and elasticity of skin reduces and fine wrinkles appear in the skin [54]. One of the common issues surrounding older adults' use of technology is the notion of "digital divide" [9, 51] which explores the idea that the older adult population are not tech-savvy and

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find it difficult to understand and use new technologies. Consequently, older adults are treated only as passive recipients of care in technology design. However, it is important to acknowledge that technologies are also not designed keeping the needs of older adults in mind.

2 RELATED WORK

Researchers have developed various digital systems to tackle issues of social isolation. Neves et al. [5] developed an iPad-based app to support asynchronous communication (through exchange of texts, audio, photos, videos) of older adults with their family and friends. Garattini et al. used the Building Bridges system, a touch-screen computer system to foster communication and reduce isolation [25]. Video games, such as playing Wii, have also been developed to stimulate social interaction, and reduce isolation [33]. Czaja et al. [16] developed a software application, PRISMS (Personal Reminder Information and Social Management System) for older adults to support a wide array of features including social connectivity [15].

However, existing literature emphasizes the need for making communication systems more accessible for this age group [3, 25]. In case of social media, there is a longitudinal study [7] that states loneliness among older adults increased from using social networking sites, because of anxiety from poor engagement. The author suggests the idea that only socializers benefiting from social media. Furthermore, the needs of the “oldest old” or frail older adults are particularly not considered in the design, deployment, and study of such systems [50]. Also, access to Internet, computers, smartphones, and other smart devices is still limited to population with higher income, education, and economic status [14].

Consequently, the adoption of ICTs and digital technologies has been slow among the aging population. Common reasons include inaccessibility, usability issues, security concerns, or high costs [10, 26, 32, 59]. Spreicer suggests that biological aging leads to reduced physical and functional capabilities, and cognitive aging reduces information processing abilities making it more difficult to learn and adapt new technologies, such as screen-based interactions [59]. Older adults avoid technologies they believe are too difficult to learn [34]. Owing to barriers stemming from limited experience of using new technologies, older adults often find digital interfaces non-intuitive and complex [17, 59]. Tangible technologies provide certain advantages as alternative to digital screen-based interfaces by providing familiar interfaces, leveraging existing capabilities and practices, thereby making it easy to learn without any training or prior knowledge of the systems [20, 32, 58, 59].

There has been some relevant research so far to address the challenges of traditional screen-based options with tangible solutions for older adults. Joshi et al. [32] developed GLiMT, a tangible alternative in the form of a wooden cube to assist older adults who struggled with traditional screen-based solutions in a care home. GLiMT opened up new and promising opportunities for interaction because it leveraged familiar inputs and movements resulting in a less complex, more intuitive option for the elderly. Researchers believe “non-tech” appearance of a system can improve acceptability among the aging population [63]. Smart pillboxes for medication management [19, 38] also leverages familiarity in design. Varesano

et al. introduced PALLA, a wireless spherical input device equipped with sensors, but without any buttons or switches so that it looks and can be used like any other non-technological everyday object [63]. PALLA was well-accepted and participants found PALLA to be amusing and engaging. While these systems were not meant for social connection, they demonstrated the potential of tangible interfaces in offering more effective technological solutions for older adults.

Early work that sparked interest in tangible technologies for social connection include papers by Mynatt et al. [47] on Digital Family Portrait and Hindus et al. [27] on social communication devices at home. Researchers have investigated the use of electronic systems to help older adults to tackle the issue of isolation by sharing well-being information. These physical electronic systems are user-facing systems that require passive participation through one-way communication. CareNet [12], another one-way communication system, offered an ambient display to adult children of an older adult to monitor various activities and events (outings, falls, medication, etc.) of the older adult. The ABRA system [53] allowed family to receive well-being data of their senior member through physiological and motion sensor while also supporting communication.

However, one-way communication systems do not include older adults as active users of the technology and only consider them as passive recipients of care. Researchers have explored technologies such as the Presence Clock [29], that allowed two-way information sharing. This system connected older adults and their caregivers and required active participation from older adults, thereby enabling them to have more control and independence. Other research works on encouraging social engagement include Social Yoga Mats [48] or games to support inter-generational play [62]. Messaging Kettle, a smart communicating device by Soro et al. [58] serves as another good example of such systems that leverage traditional routine of making tea to improve social connectedness among the aging population.

3 RESEARCH QUESTIONS

Ishii and Ullmer [31] introduced the idea of tangible user interfaces 20 years back. But existing research indicate the lack of evidence in research in using tangible technologies to improve social connection among older adults. There is a need for more integrated work to address issues pertaining to social isolation among older adults. Addressing this gap in literature, my dissertation work aims (1) to explore the needs and challenges of older adults when it comes to checking-in with friends and family (2) to investigate the potential of tangible technologies in providing an efficient solution in supporting older adults to check-in on their loved ones and maintain social connection (3) to suggest design considerations for HCI researchers for designing tangible technologies for older adults with the aim to address the issue of social isolation.

In my doctoral study, I address the following research questions:

- (1) RQ1: Who are a part of older adults’ social circle and what is the current check-in behavior of older adults?
- (2) RQ2: How can check-in systems help improve social connection for older adults?

- (3) RQ3: What affordances of tangible technologies make them suitable for older adults?

4 METHODOLOGY

The research questions will be answered through the studies I have conducted and the study I plan to conduct in future as part of my dissertation:

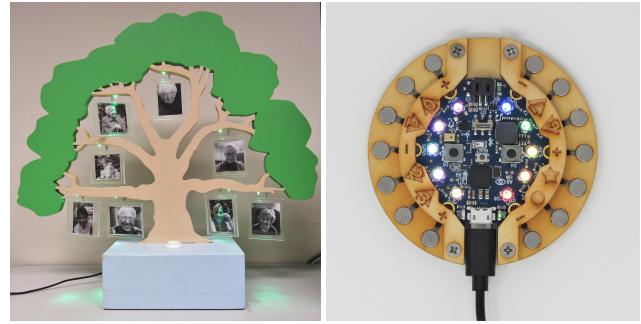
- Conducted a diary study with 16 older adults to investigate their current check-in behavior (RQ1).
- Worked on a check-in system and conducted a field deployment at 16 older adult houses to gather feedback on usability and utility (RQ2).
- Based on results from the previous study, developed another novel check-in system, which is an advanced version of the previous check-in system (RQ2).
- Plan to build a framework with design consideration for HCI researchers developing tangible systems for older adults (RQ3).

4.1 Diary Study

We have conducted a diary study (RQ1) with 16 older adults to investigate their current check-in behavior. We asked them to log their daily communication with loved ones (i.e., family, friends, neighbors, etc.) on a paper-based, physical diary. It was observed that although participants listed family among their strongest relationship ties, friends formed an integral part of an older adults' social circle when it came to daily communication. There was a significant difference between their perceived check-in behavior with their actual check-in behavior. Looking at the frequency of communication with friends for every individual participant, we found that 75% of the participants had up to third-quarters of their communication with their friends as compared to all the other relationships. The implication from this work was to shift the focus from family to connecting older adults with their friends/peers. Through this study we provide evidence for the same, and encourage the research community to design systems that broaden social networks, thereby providing opportunities to redistribute caregiving and reduce the burden on a single entity in an older adults' social network.

4.2 Check-in Tree - Field Deployment

Consequently, I worked on an IoT system, a Check-in Tree [Fig. 1a], a peer-based physical check-in system to connect 8 older adults and their peers, across the globe [2]. When an older adult wakes up in the morning and presses the button at the base of the tree, the led associated with that older adult turns on, and every other older adult in the peer group gets notified, thereby facilitating a morning check-in. Results from the two-week field deployment study (RQ2) with 16 older adults showed that older adults found the technology very intuitive and easy to use. Despite zero training, participants were able to use and understand the system without any issue. All participants engaged with the system and found it a fun thing to do. The most significant finding from the study was that it showed a sense of bonding developing even among strangers, and started showing concerns for each other when someone was not checking-in within a certain time. Participants also found the



(a) Check-in Tree

(b) Check-in Toolkit

Figure 1: Tangible peer-based check-in systems

technology non-stigmatizing since they were actively checking-in on others instead of being only on the receiving end of care, or being passively monitored.

4.3 Check-in Toolkit - System Overview

The field deployment study revealed that participants found the functionality of the Check-in Tree to be very useful, but were not comfortable with the size and weight of the Tree. They found it clunky and it took up desk space. An interesting observation was the varied range of alternate solutions the participants suggested. All of them wanted the functionality to be the same, but the form factor different. Consequently, I developed a novel electronic system, the Check-in Toolkit [Fig. 1b] (RQ2), built using an Adafruit Bluefruit, which simulated the functionality of the Tree, except that this toolkit is completely customizable. The motivation was to build a customizable version of the Check-in Tree to add scalability and cater to the aesthetic needs of older adults. This Toolkit could be attached to any surface or sewn into any fabric, turning any everyday object into a user interface, thereby eliminating the rigidity in user interface design. In past literature as well, researchers have found that older adults find technology more meaningful if they find value or personal relevance in using it [11, 23, 44]. Older adults have abandoned technologies that do not cater to their specific needs and sense of aesthetics [4, 11]. They value technology that are flexible enough to allow older adults to personalize and contextualize devices [42]. Customization is therefore a valuable feature that can offer tailored solutions to older adults. Building one's own artifact increases the value they place on it [52]. The technical architecture depicted in Fig. 2 for both the check-in systems shows the use of publish-subscribe protocol by RabbitMQ broker in publishing notifications from one participant to all subscribed participants in the peer-group simultaneously.

5 PROPOSED STUDY AND EVALUATION

I propose a study that would potentially extend my past work and answer my third research question, RQ3. I would like to conduct a co-design workshop with older adults using the Check-in Toolkit. The workshop will aim to generate ideas and create design concepts for the user interface of the Check-in Toolkit. The study will help us understand a) older adult participants' preferences of user

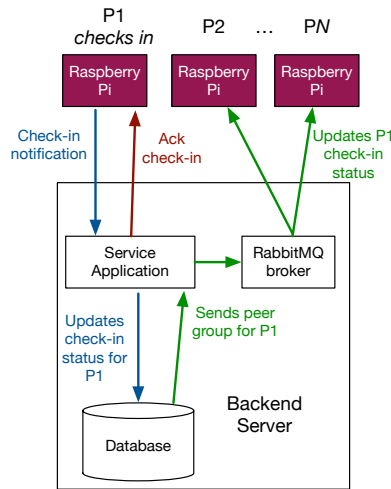


Figure 2: Backend Architecture updating check-in status for Participant 1 (P1) for both Check-in Tree and Check-in Toolkit.

interfaces for a tangible check-in system, b) the utility (if any) of offering the agency to choose their own preferred user interfaces, and consequently c) their acceptance of customizable check-in systems for checking-in with their peers. Finally, based on the results from this study and the study with the Check-in Tree, I aim to develop a framework with design considerations for HCI researchers interested in developing tangible systems for older adults. I aim to discuss the affordances that tangible systems can provide as an alternative to digital screen-based interfaces for older adults.

6 (EXPECTED) CONTRIBUTIONS

Through my dissertation, I have investigated the issue of social isolation among older adults and explored the utility of tangible check-in systems for improving social connection among older adults. This dissertation aims to make the following contributions -

- Explores the benefits of everyday check-in for older adults and their peers in reducing isolation by making them active participants in the check-in process.
- Investigates the potential of tangible systems in delivering intuitive and familiar interfaces, which in turn reduces the learning curve and eliminates the necessity for training, saving time, money, and effort invested in training older adults.
- Examines the benefits of enabling customization in a check-in system, allowing individuals to have agency over their choice of user interfaces. This addresses aesthetic needs and ensures seamless integration into their surroundings. Moreover, these interfaces can be updated, modified, and further customized to adapt to changing preferences over time.
- Provides a framework with design consideration to inform HCI researchers on better design of tangible technologies that cater to the specific needs of older adults, thereby resulting in better technology adoption.

Throughout my research, my primary focus has been on addressing the issues of social isolation among older adults by providing everyday check-in solutions for improving social connection. I have dedicated my efforts to exploring the design and development of tangible technologies that can effectively support this population. Additionally, I strive to address issues of bias and ageism in technology design, aiming to create inclusive solutions that enable older adults to stay socially connected and maintain their independence for an extended period.

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7 BIOGRAPHICAL SKETCH

Pallabi Bhowmick is a PhD Candidate in the Department of Informatics (Health Informatics track) at Indiana University Bloomington. Her research focuses on improving social connection among isolated older adults through tangible technologies. Her primary advisor is Prof. Erik Stolterman Bergqvist, who is also affiliated to Indiana University Bloomington as Senior Executive Associate Dean at the Luddy School of Informatics, Computing, and Engineering. Pallabi Bhowmick has passed her Qualifying Examination, defended her Dissertation Proposal, and is expected to complete her PhD by May 2024. She has a clear research direction and is therefore, at the ideal stage to benefit from attending the Doctoral Colloquium at UbiComp/ISWC 2023. She solicits feedback from the UbiComp community on the use of peer-based networked check-in systems for improving social connection and thereby, facilitate better mental health and well-being of older adults.

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